Predicting Missing Intervals in CGM Data

Xinyu Dong

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Background

- Continuous Glucose Monitoring (CGM) data are widely used in diabetes research
- Provide dense time series of glucose levels
- However:
  - Data often contain **missing intervals**
- Goal of this work:

  Predict when missing intervals begin and how long they last

Why Missingness Matters in CGM

- CGM provides **high-frequency glucose measurements**
  - typically every 5–15 minutes
- Missing intervals arise due to:
  - sensor failure or detachment
  - calibration issues
  - physiological artifacts
- Consequences:
  - biased estimates of glucose variability
  - incorrect clinical summaries (e.g., time-in-range)
  - degraded performance of downstream models
- Key insight:
  - Missingness is **informative**, not random

Problem Formulation

- Binary outcome:

$$y_{\text{onset}} = \begin{cases} 1 & \text{start of missing interval} \\ 0 & \text{otherwise} \end{cases}$$

- Continuous outcome:

$$\text{duration (minutes)}$$

- Extremely rare event:
  - ≈ 216 positives out of $\approx 25{,}000$

Methodology Overview

- Two-stage modeling:
  1. Predict onset probability
  2. Predict duration conditional on onset
- Compared two approaches:
  - GLM baseline
  - Bayesian MCMC model
- Evaluation:
  - 5-fold grouped cross-validation
  - Metrics: AUROC, AUPRC, RMSE

GLM Baseline

Onset model:

$$\text{logit}\big(P(y_{\text{onset}} = 1 \mid X)\big) = X^\top \beta$$

- Standard logistic regression
- Trained on highly imbalanced data

Duration model:

$$\text{duration} = X^\top \gamma + \epsilon$$

- Fit only on observed missing intervals
- Limited sample size

Bayesian Model

Onset:

- Logistic model with priors:

$$\beta \sim \mathcal{N}(0, \sigma^2 I)$$

- Posterior inference via MCMC

Key advantage:

- Regularization via priors
- More stable under rare-event settings

Prediction:

$$\hat{p}_i = \mathbb{E}[P(y_i = 1 \mid X_i, \beta)]$$

Cross-Validation Design

- 5-fold grouped CV (by patient)
- Prevents leakage across time/subjects
- For each fold:
  - Train on 80%
  - Evaluate on 20%
- Metrics:
  - AUROC (ranking ability)
  - AUPRC (rare-event performance)
  - RMSE (duration)

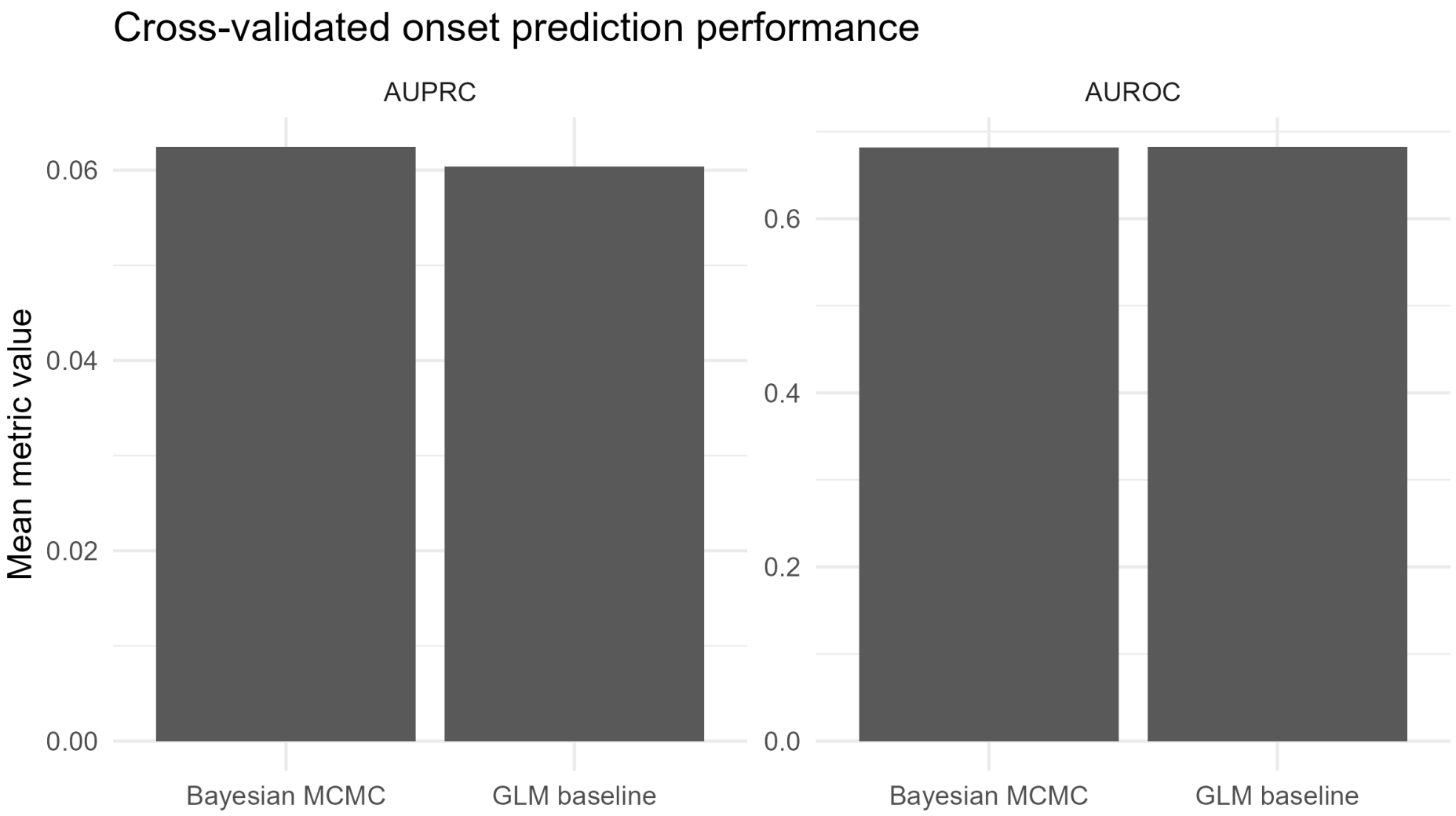
Key Challenge: Rare Events

- Prevalence $\approx 0.05\%$
- Implications:
  - AUROC can appear high even with poor precision
  - AUPRC is more informative
- Model goal:
  - Not perfect classification
  - **Correct ranking of high-risk windows**

Cross-Validated Performance

- Onset prediction shows **moderate discrimination**:
  - AUROC ranges from ~ 0.65 to 0.90
- However, precision remains low:
  - AUPRC is close to 0 in several folds
  - Reflects **extreme class imbalance** ($< 0.1\%$ positives)
- Key interpretation:
  - Model can **rank high-risk windows**
  - But cannot reliably identify positives directly

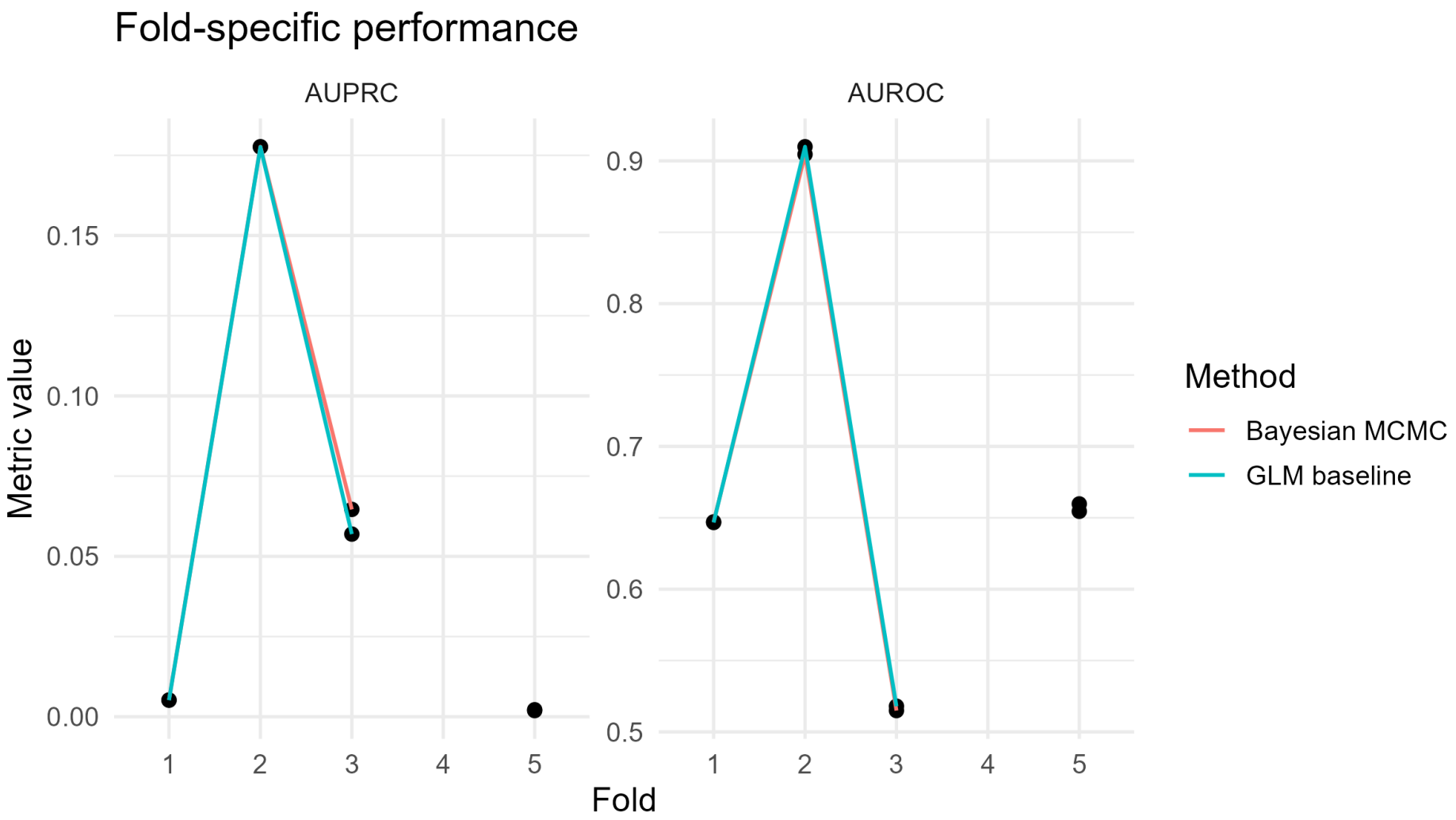
Cross-Validated Performance



Performance Variability Across Folds

- Substantial variability across folds:
  - Some folds achieve high AUROC (~ 0.9)
  - Others are close to random
- Explanation:
  - Only $\sim$216 total events
  - Some folds contain **very few or no positives**
- Implication:
  - Performance is **data-limited**, not just model-limited

Performance Variability Across Folds

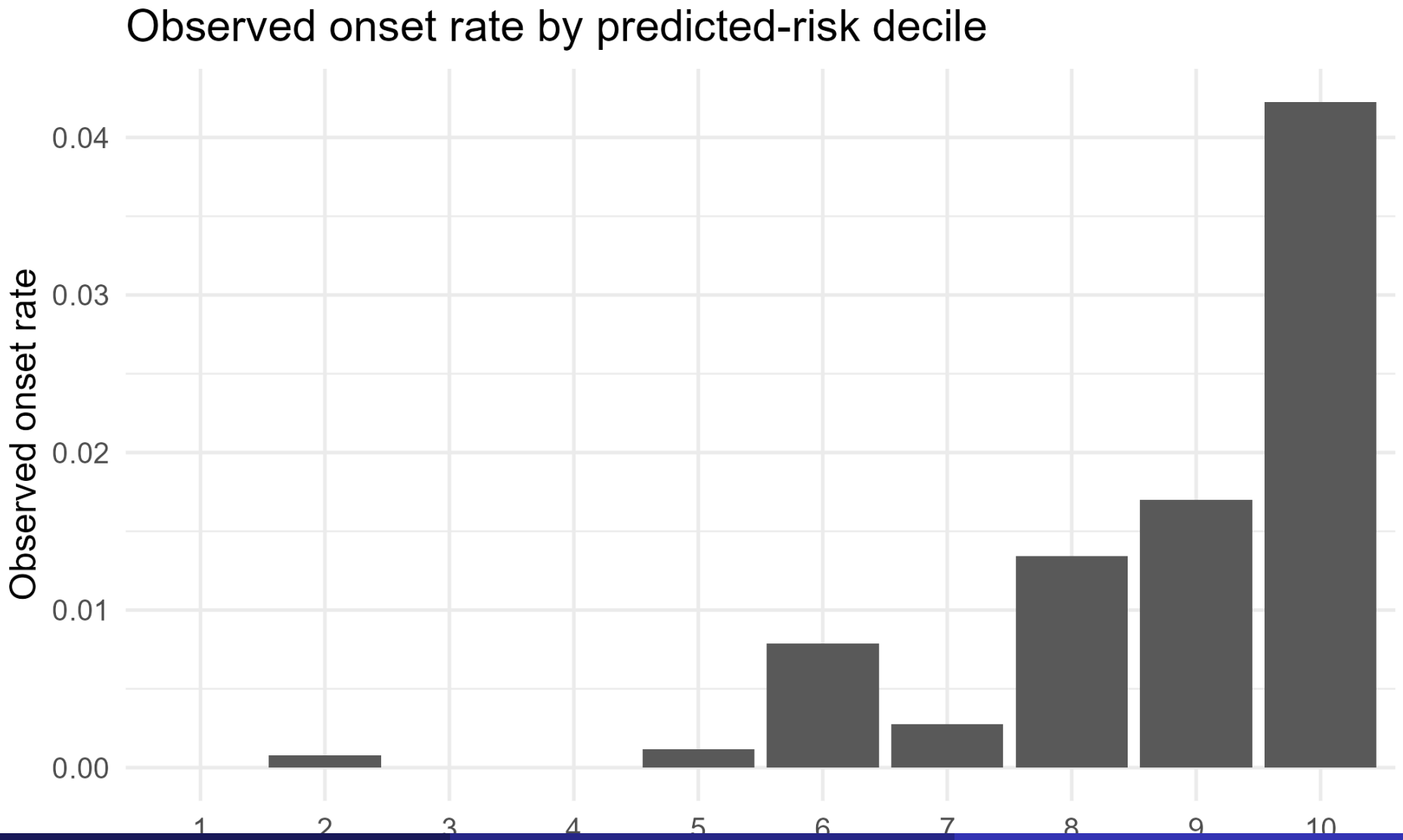


GLM vs Bayesian Model

- Both models achieve **similar AUROC and AUPRC**
- Differences:
  - Bayesian model:
    - Slightly more stable across folds
    - Better calibrated probabilities
  - GLM:
    - Simpler and faster
    - Comparable discrimination
- Key takeaway:
  - **Model choice matters less than data scarcity**

Risk Stratification Performance

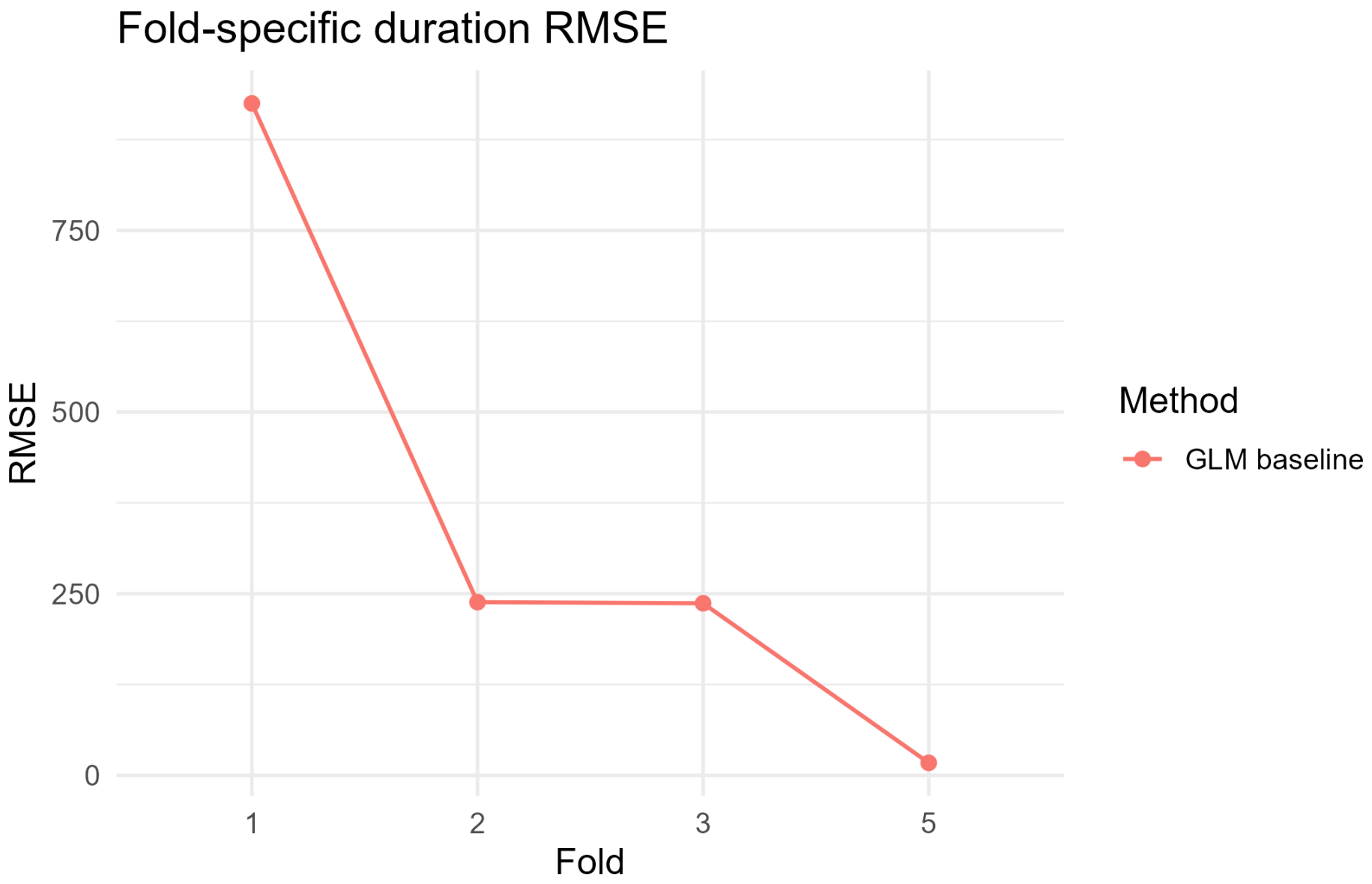
- Evaluate model via **risk deciles**
- Compare predicted vs observed onset rates



Duration Prediction Results

- Duration model shows **poor and unstable performance**
- Evidence:
  - RMSE varies widely across folds
  - Predictions highly dispersed
- Reasons:
  - Only 216 training examples
  - Duration has high intrinsic variability
- Conclusion:
  - Duration prediction is **not reliable in current setting**

Duration Prediction Results



Interpretation

- Onset prediction is **feasible**
- Duration prediction is **unstable**
- Key takeaway:
  - Even under extreme imbalance,
  - models can provide useful **risk ranking**

Discussion

- In rare-event prediction:
  - Should we prioritize AUPRC over AUROC?
- What metric best reflects practical usefulness?

Thank You!